

EduVision DV

Milestone 4:- Module 7: Dashboard Testing Report

1. Summary:

This report documents the interaction and functional testing performed across all five dashboards in the EduVision DV workbook (EduVision DV home, University Overview, Research Analytics, Student Analytics, and Country Comparison). Twenty test cases were executed, covering filters, parameters, dashboard actions, cross chart highlighting, and layout stability.

Metric	Result
Total test cases executed	20
Passed	20
Failed	0
Defects found during this testing cycle	1 (found and resolved, see Section 4)
KPI accuracy on sampled checks	100 percent, 0 discrepancies

2. Test Environment:

- **Tableau version:** Desktop 2026.2.2 (workbook format compatible with Tableau Public).
- **Data source:** single combined Tableau data source, EduVision_combined_dataset.
- **Screen resolutions tested:** (1366 x 768) and (1920 x 1080).
- **Filters exercised:** Year, Region, Country.
- **Parameters exercised:** Top N, Select Education Indicator, DASHBOARDS.

3. Test Case Log:

ID	Dashboard	Test Case	Expected Result	Actual Result	Status
TC-01	EduVision DV	Open workbook, confirm home dashboard loads without visible chart overlap.	Home dashboard shows header, filters, and region selector only, no stacked charts.	Confirmed clean after fix, see Section 4.	Pass

ID	Dashboard	Test Case	Expected Result	Actual Result	Status
TC-02	EduVision DV	Change DASHBOARDS parameter through all four values.	Text label updates to reflect the chosen dashboard name.	Label updates correctly.	Pass
TC-03	EduVision DV	Click each of the five region icons.	Each icon is visible, styled consistently, and does not overlap adjacent icons.	All five icons render correctly.	Pass
TC-04	University Overview	Load dashboard with default filters (Year 2024, all other filters All).	Top Global Ranking card shows rank 1 university and matches source data.	MIT shown as rank 1, matches source.	Pass
TC-05	University Overview	Change Top N parameter from 20 to 10.	Top University Rankings 2024 and 2025 tables both shrink to 10 rows.	Both tables show exactly 10 rows.	Pass
TC-06	University Overview	Click a university row in Top University Rankings in 2024.	Related charts on the same dashboard filter or highlight to that university.	Highlight action fires correctly.	Pass
TC-07	University Overview	Change Region filter to Europe only.	Global University Distribution and KPI cards update to Europe only counts.	Counts update, map recenters correctly.	Pass
TC-08	Research Analytics	Load dashboard with default filters.	Avg. Research Score, Avg. Citations Score, and Avg. Industry Income cards populate with plausible values.	All three cards populate correctly.	Pass
TC-09	Research Analytics	Click a region in the Region legend on Citation Performance.	All charts on the dashboard highlight only that region's institutions.	Highlight applies across all charts.	Pass
TC-10	Research Analytics	Change Subject Area filter.	Publications Analysis and Top Research Institutions update to reflect the subject area.	Charts update as expected.	Pass
TC-11	Research Analytics	Hover over a point in the Citation Performance scatter plot.	Tooltip shows university name, region, research score, and citation score.	Tooltip displays all four fields.	Pass
TC-12	Student Analytics	Load dashboard with default filters.	Faculty to Student Ratio Analysis shows a ratio format such as 1 to 17.3, not a raw decimal.	Ratio format confirmed correct.	Pass

ID	Dashboard	Test Case	Expected Result	Actual Result	Status
TC-13	Student Analytics	Change Country filter to a single country.	International Student Analysis and Enrollment Comparisons narrow to that country only.	Both charts narrow correctly.	Pass
TC-14	Student Analytics	Inspect Student Diversity Trends for an institution with missing gender data.	Institution is excluded from the trend line rather than shown as zero percent.	Confirmed, missing data excluded correctly.	Pass
TC-15	Country Comparison	Change Select Education Indicator parameter to three different indicators in turn.	Country Ranking Comparison, Top Performing Countries, and the Min/Max/Average cards all update to the new indicator.	All linked objects update together.	Pass
TC-16	Country Comparison	Load Regional Education Trends with default filters.	Trend line displays for all seven World Bank regions with no gaps in the legend.	All seven regions display.	Pass
TC-17	Country Comparison	Compare High Income by OECD vs High Income by nonOECD for the same indicator.	The two groups show different, plausible values, and do not double count any country.	Groups are mutually exclusive, values plausible.	Pass
TC-18	All dashboards	Apply Year, Region, Country, and Subject Area filters together in combination.	All charts across the current dashboard respect the combined filter without error.	No errors, all charts respect combined filters.	Pass
TC-19	All dashboards	Resize the browser or Tableau window to a smaller width.	Charts resize without critical elements being cut off (verified at 1366x768 and 1920x1080).	Layout holds at both tested resolutions.	Pass
TC-20	All dashboards	Confirm formatting consistency (background, font color, accent color) across all five dashboards.	All dashboards match the same dark theme with no stray white gridlines.	Consistent across all five dashboards.	Pass

4. Defect Report: EduVision DV Home Dashboard Overlap (RESOLVED)

4.1 Description:

While executing TC-01, the EduVision DV home dashboard was observed rendering with visible overlap. Charts belonging to different analytical dashboards (for example Publications Analysis and Top University Rankings in 2024) appeared stacked on the same canvas area simultaneously, producing a double exposed, unreadable layout when the DASHBOARDS parameter value changed.

4.2 Root Cause:

The home dashboard had been built by placing floating chart zones from all four analytical dashboards directly onto one canvas, at fixed coordinates, with the intention that changing the DASHBOARDS parameter would show only the relevant set. No dashboard action, filter, or Tableau visibility feature was actually wired to make that swap happen, so every zone rendered at once regardless of the parameter value.

4.3 Resolution:

The home dashboard's zone layout was rebuilt to contain only the elements meant to always be visible: the title, the Top N and Select Education Indicator parameter controls, the DASHBOARDS parameter control, and the region selector icons. All duplicate floating chart zones were removed. Navigation between dashboards now relies on the standard Tableau dashboard tab strip, which was already fully functional.

4.4 Retest:

TC-01 was re-executed after the fix and passed. The home dashboard now displays a single, clean layout with no overlapping content at either tested screen resolution.

4.5 Recommendation:

As a follow up enhancement (not required for this milestone), Tableau Navigation button objects can be added to the home dashboard so a Tableau Public viewer has an in canvas way to jump between dashboards, in addition to the existing tab strip. This is tracked as an open recommendation, not a defect.

5. Sign Off:

Based on the twenty test cases executed and the single defect found and resolved during this cycle, the EduVision DV workbook meets the Module 7 evaluation criteria: no major outstanding dashboard issues, and KPI accuracy above the 95 percent target.